6.A Thermochemistry I

The Nature of Energy. When it comes to energy, the general feeling is often that “we know it when we see it.” Unfortunately, that vague sense isn’t good enough for scientists, who like to define concepts precisely and, wherever possible, quantitatively. Here, we will offer up a precise definition for energy and examine the most important types of energy for chemistry.

- **Thermochemistry** is the study of heat and energy in chemical reactions.

- **Energy** is the capacity of a thermodynamic system (more on that soon) to do work or transfer heat.

- **Mechanical energy** is due to the motion or position of an object.

  - **Kinetic energy** is energy of motion. For a system or particle in motion with mass *m* and velocity *v*, the kinetic energy is

$$KE = \frac{1}{2}mv^2$$

  - **Potential energy** is energy related to the relative position of a particle or system. Examples include gravitational and electrical potential energy. Potential energy is often inversely related to the distance or relative position *r*,

$$PE \propto r^{-1}$$

- The SI unit of work, heat, and energy is the **Joule (J)**.
  One Joule is equivalent to 1 kg·m^2/s^2.

  - 1 liter-atmosphere is equal to 101.325 Joules.

  - 1 calorie is equal to 4.184 Joules.

  - 1 Calorie (nutritional calorie) is equal to 1000 calories or 4184 Joules.

Thermodynamic Systems. In chemical thermodynamics, we are interested in the properties of thermodynamic systems, which are portions of the physical world with a well-defined boundary in an internal state of equilibrium. Properly defining system and surroundings (everything else in the "universe") is a key first step in solving problems in thermodynamics. Keeping this in mind will serve you well in more advanced thermodynamics courses, too!

- Proper definitions of the system, surroundings, and universe are key to solving problems in thermodynamics.
    - The **system** is a subdivision of the universe with a well-defined boundary, existing in a state of internal equilibrium (with no tendency toward change on the macroscopic scale).
    - The **surroundings** are everything in the universe outside of the system.
    - The **universe** is not simply everything there is, but everything we care about. Assumptions often idealize the universe or "cut off" consideration at a certain distance from the system.

system + surroundings = universe

- Thermodynamic systems are classified according to their ability to transfer matter and energy in or out.
    - **Open system:** both matter and energy can be exchanged between the system and the surroundings (transfer of matter implies transfer of energy)
    - **Closed system:** energy—but *not* matter—can be exchanged between the system and the surroundings
    - **Isolated system:** neither energy nor matter can be exchanged between the system and the surroundings

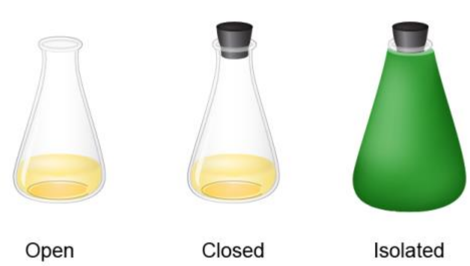


Figure 6.3. Examples of open, closed, and isolated systems.

Energy Transfers, Heat, and Work. We don't "see" energy until it is transferred. Our definition above implies that energy can be transferred in only two ways: via the transfer of heat or via work. Here, we'll see that there is a very simple and tautological reason for this. Heat and work differ only in the extent of their randomness, and heat, work, and energy all share the same units.

- **Work** is the result of a force acting through a distance—a kind of energy transfer in which the particles of the system undergo coordinated motion in a nonrandom direction.

- **Heat** is the result of a transfer of thermal energy—a kind of energy transfer in which the particles of the system undergo random motion.

- In chemistry, the sign of work is defined with reference to energy transfer to or form the system.
  - Work done **on** the system by the surroundings is **positive**.
  - Work done **by** the system on the surroundings is **negative**.

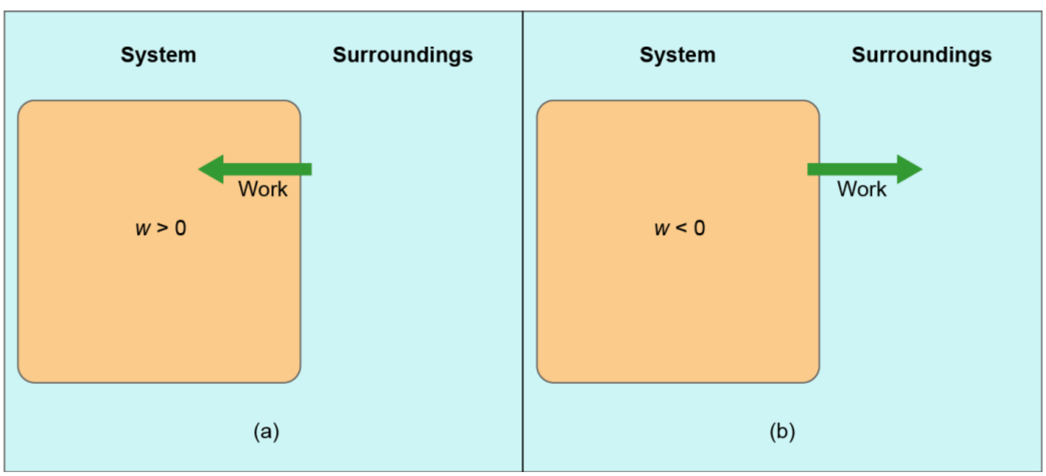


Figure 6.4. Work done on the system is positive; work done by the system is negative.

- The sign of heat is also defined with reference to energy transfer to or from the system.
    - Heat (thermal energy) transferred **to** the system from surroundings is **positive**.
    - Heat (thermal energy) transferred **from** the system to surroundings is **negative**.

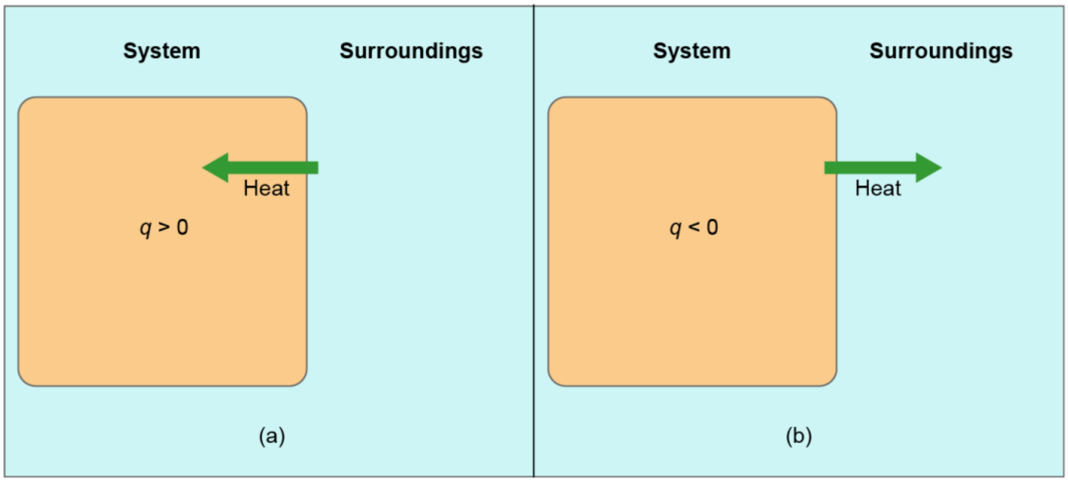


Figure 6.5. Heat transferred to the system is positive; heat transferred from the system is negative.

6.1. Is work done on the system or by the system in the following scenarios?
In each case, the system is bolded.

1. A **person** rows a boat across a lake.
    A. Work is done on the system ($w > 0$).
    B. Work is done by the system ($w < 0$).

2. A person pedals a **bicycle** down the street.
    A. Work is done on the system ($w > 0$).
    B. Work is done by the system ($w < 0$).

The First Law of Thermodynamics. You've probably heard the first law of thermodynamics expressed before as the idea that "the total energy of the universe is constant," or perhaps "energy is neither created nor destroyed." While these formulations will remain important for us in chemical thermodynamics, we will more commonly apply the related idea that energy transfers to or from a thermodynamic system are entirely accounted for by heat and work.

- **Internal energy (*U* or *E*)** is the sum of the kinetic and potential energies of all particles within a thermodynamic system.

- **The first law of thermodynamics** establishes fundamental constraints on energy and energy transfers.
    - The total energy of the universe is constant.
    - Energy is neither created nor destroyed.
    - All energy transfers (changes in internal energy) are due to heat and/or work.

- A change in internal energy is denoted ΔU…

$$\Delta = \text{final} - \text{initial}$$

$$\Delta U = U_{fin} - U_{init}$$

- All energy transfers (changes in internal energy) are due to heat and/or work.
  We can express this idea mathematically and will frequently use this equation as a starting point when solving problems in thermochemistry.

$$\Delta U = q + w$$

- Recall the sign conventions for heat and work. The convenience of the way we think about the sign of work *w* should now be apparent!

Parameter	Sign	Meaning
q	+	Heat flows into the system.
q	–	Heat flows out of the system.
w	+	Work is done on the system.
w	–	Work is done by the system.
Δ*U*	+	The system gains internal energy.
Δ*U*	–	The system loses internal energy.

Table 6.2. Sign conventions for heat, work, and internal energy change.

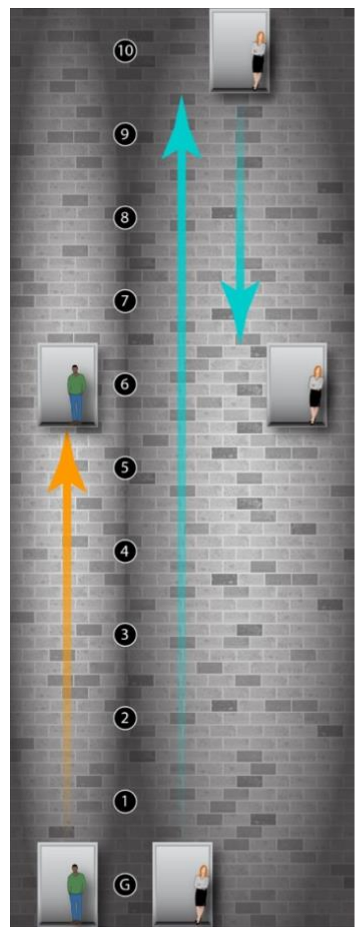
6.2. For a particular process $q = -10$ kJ and $w = +25$ kJ. Which statement is *true*?

A. Heat flows from the surroundings to the system.
B. The system does work on the surroundings.
C. ΔU is equal to –35 kJ.
D. All these statements are true.
E. None of these statements is true.

State and Path Variables and Functions. So far in our study of chemistry, we have considered several variables of relevance to thermodynamics. Heat q, work w, and internal energy U obviously come to mind, but temperature T, volume V, and number of moles n are three others. All these variables (and functions derived therefrom) fall into two fundamentally different classes: state variables and path variables. Although state and path variables are frequently juxtaposed in introductory chemistry,[1] they are fundamentally different things and are dealt with in very different ways. Remember this!

- **State variables (functions)** have values that depend only on the internal state of the system—the positions and velocities of all particles.
  - Well defined for an unchanging thermodynamic system
  - Changes in state variables are easy to measure and calculate:

$$\Delta X = X_2 - X_1$$



  - A "state function" is just some mathematical expression that includes only state variables.
- Internal energy is one example of a state variable.
- You may already be familiar with many state variables: volume, mass, temperature, number of moles, pressure, …
- We will encounter other important state variables and functions later in this chapter and in chapter 18.
- State variables are denoted with upper-case letters. (The exception is n for moles; N is used for the absolute count of molecules in the system.)

[1] Specifically, path variables are often reduced or simplified to changes in state variables. In later thermodynamics courses, when your knowledge of calculus will be assumed, this will stop being the case.

- **Path variables (functions)** are associated with processes and change; their values depend in general on the intimate details of the process.
    - State variables "belong" to a state of an unchanging system.
      Path variables "belong" to a process in a changing system.
    - Path variables are frequently difficult to calculate. We generally must integrate over a path from an initial state S_1 to a final state S_2. For heat and work,

$$q = \int_{S_1}^{S_2} C(T)\, dT \qquad\qquad w = -\int_{S_1}^{S_2} P(V)\, dV$$

- When C does not vary with T and P does not vary with V, the path variables q and w reduce to changes in state functions ΔT and ΔV. Yay! This will be the situation throughout CHEM 1211K and 1212K.

$$q = \int_{T_1}^{T_2} C\, dT = C(T_2 - T_1) = C\,\Delta T$$

$$w = -\int_{V_1}^{V_2} P\, dV = -P(V_2 - V_1) = -P\,\Delta V$$

- Consider 1 mole of liquid ethanol at 20 °C and atmospheric pressure.

    - What do we know about this sample from just this information? The **state variables:** *P*, *T*, *n*, *U*, etc.

    - The ethanol could have reached this state in different ways; each is associated with different values for the **path variables** *q* and *w*. For example,
        - Solid ethanol at –120 °C is heated directly to 20 °C.
        - Solid ethanol at –120 °C is heated first to 50 °C and then cooled to 20 °C.

    - The change in internal energy (ΔU) is the same in both cases, but *q* and *w* differ in general for the two different paths.

A Deeper Look at Work. Here, we dive deeper into work done on or by gases, which are frequently systems of interest in chemical thermodynamics.

- Work is about more than just pushing macroscopic objects against friction or gravity.

- The work done on a gas to compress it (or by the gas as it expands) is called **pressure-volume (*PV*) work**.

- *PV* work can be expressed in units based on the product of a pressure and volume, such as **liter-atmospheres (L·atm)**.

$$w = -P\,\Delta V$$

6.3. The change in internal energy for a given system is –325 J. This system loses 155 J of energy as heat as it expands from an initial volume of 15.0 L at 1.00 atm of pressure. What is the final volume of the system in liters?

One liter-atmosphere is equivalent to 101.3 Joules; one Joule is equivalent to 0.00987 liter-atmospheres.

A Deeper Look at Heat. We saw above that heat is calculated as a temperature change times a quantity *C*, which we'll call heat capacity, that relates heat and temperature. An important moral of this story is that heat and temperature are not the same: heat is a path variable (related to a temperature *change* or transfer of thermal energy) while temperature is a state variable (related to the average kinetic energy of particles in the system).

- Heat and temperature are not the same. Heat is calculated from a change in temperature using the heat capacity of the system—or some intensive measure of heat capacity times either the mass or amount in moles of the system.

- The amount of heat required to raise the temperature of 1 gram of a substance by 1 degree Celsius is called the **specific heat (*c*)** of the substance. Heat, specific heat, mass, and temperature change are related:

$$q = m \cdot c \cdot \Delta T \qquad\qquad c = \frac{q}{m \cdot \Delta T}$$

- The **molar heat capacity (*C_m*)** of a substance is the heat required to raise the temperature of 1 mole of the substance by 1 degree Celsius. Heat, molar heat capacity, amount in moles, and temperature change are related:

$$q = n \cdot C_m \cdot \Delta T \qquad\qquad C_m = \frac{q}{n \cdot \Delta T}$$

- Specific heat and molar heat capacity are intensive properties; **heat capacity (*C*)** itself is extensive.

$$q = C \cdot \Delta T \qquad\qquad C = \frac{q}{\Delta T}$$

- Note that these equations only hold in the absence of a phase change. We will address phase changes later in the course.

Substance	Specific Heat (J g^{-1} $^oC^{-1}$)	Substance	Specific Heat (J g^{-1} $^oC^{-1}$)
CO_2	0.852	Al	0.892
CO	1.04	Cr	0.45
H_2	14.4	Co	0.46
N_2	1.04	Cu	0.385
O_2	0.922	Au	0.129
$H_2O(g)$	2.042	Fe	0.442
$H_2O(l)$	4.184	Pb	0.13
$H_2O(s)$	2.089	Mg	1.0
		Ag	0.24
		Na	0.293
		Sn	0.22
		Zn	0.388

Table 6.3. Specific heats of select substances.

6.4. What is the specific heat capacity of a substance if 1155 J of heat are needed to raise the temperature of 0.100 kg of the substance from 25.0 ^oC to 55.0 ^oC? Enter your response in Joules per degree Celsius per gram (J $^oC^{-1}$ g^{-1}).

6.B Thermochemistry II

Introducing Enthalpy. Chemical reactions are most frequently run under conditions of constant pressure, with either the earth's atmosphere or a blanket of inert gas enforcing a consistent pressure near 1 atmosphere. Under these circumstances, chemical reaction systems are subject to both heat and work, but we find that the heat evolved or absorbed by the system is equal to the change in a state function called enthalpy (*H*).

- Most chemical reactions are run in open systems in which both heat and work can be nonzero. Usually, the pressure is constant at or near atmospheric pressure.

- We are typically more interested in the heat than the work of chemical reactions.

- Starting from the first law, we can identify a new state function **enthalpy (*H* = *U* + *PV*)**, the change in which (Δ*H*) is equal to the heat of reaction at constant pressure.

$$\Delta U = q_P + w_P = q_P - P\,\Delta V$$

$$q_P = \Delta U + P\,\Delta V$$

$$H = U + PV \Rightarrow q_P = \Delta U + P\,\Delta V = \Delta H$$

- **Exothermic processes** involve the transfer of heat from the chemical system to the surroundings. The enthalpy change Δ*H* is negative.

- **Endothermic processes** involve the transfer of heat to the chemical system from the surroundings. The enthalpy change Δ*H* is positive.

Figure, section 6.4. Exothermic and endothermic processes:
$4\,Fe(s) + 3\,O_2(g) \rightarrow 2\,Fe_2O_3(s) + heat$
$NH_4NO_3(s) + heat \rightarrow NH_4NO_3(aq)$

- The terms "enthalpy change," "enthalpy of…," and "heat of…," are all loosely used to refer to enthalpy changes. There is no absolute zero of enthalpy.

- ΔH can be measured or, if related enthalpies are known, calculated. Either way, enthalpies of reaction are expressed as kilojoules per mole of reaction events (kJ/mol).
    - Enthalpies are measured using **calorimetry**.
    - Enthalpies are calculated by applying **Hess's law**.

Calorimetry. The essence of measuring heat involves "submerging" a system in surroundings with known heat capacity. Heat transfer due to a process in the system causes a temperature change ΔT in the surroundings, q_{surr} is calculated using the known heat capacity, and q_{sys} is equal to q_{surr} but opposite in sign, according to the first law. Writing a correct heat balance equation is the key step when solving calorimetry problems!

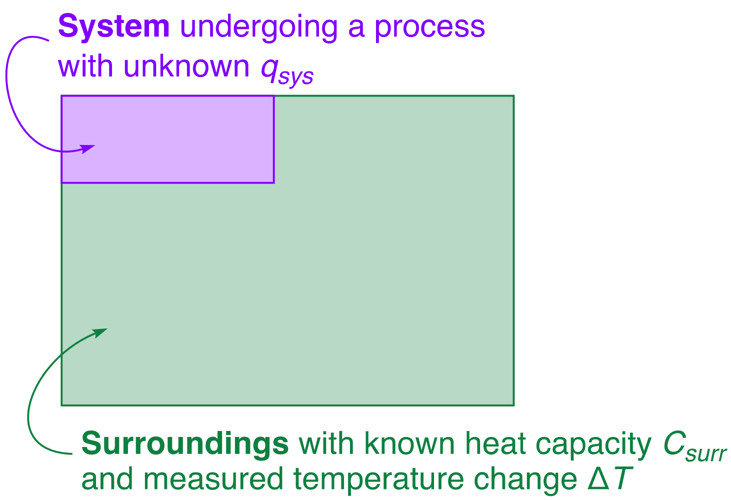
- **Calorimetry** is a laboratory technique for quantifying heat transfer by measuring temperature changes in the substances involved.

- Calorimetry calculations are based on a **heat balance** of the system and surroundings. Heat transfer to the surroundings q_{surr} is measured and q_{sys} is inferred from it.

$$\sum_i q_i = 0$$

$$q_{sys} + q_{surr} = 0$$

$$q_{sys} = -q_{surr}$$



6.5. A 25.0 g block of metal initially at 95.0 °C is placed in a calorimeter containing 121 g of water initially at 27.2 °C. The specific heat of the metal is 0.49 J °C^{-1} g^{-1}. The specific heat of water is 4.184 J °C^{-1} g^{-1}.

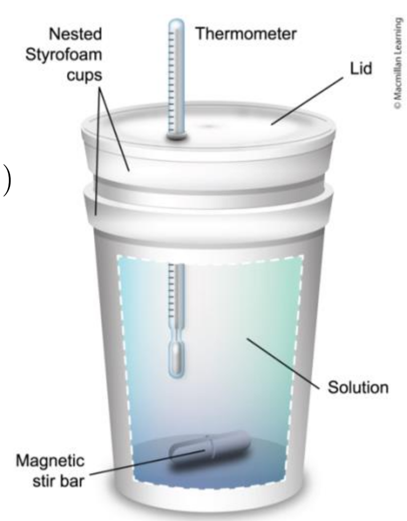
Assuming that no heat is lost to the calorimeter or surroundings, what is the final temperature of the water and metal? Enter your response in degree Celsius to the nearest 0.1 °C. Start by drawing a picture and writing a heat balance.

- The system of interest in chemical calorimetry is typically a submicroscopic chemical reaction. The heat evolved or absorbed by the system is determined by measuring the temperature change in the *surroundings*.
  - The temperature of the solution (surroundings!) is measured, not the temperature of the system itself.
  - We cannot see or directly measure the reacting molecules.

- **Constant-pressure calorimetry** involves running a reaction in an insulated vessel under atmospheric pressure. In this situation, the heat absorbed or released by the reaction (q_P) is equal to the enthalpy change (ΔH).

- The reaction is typically happening in aqueous solution.
  - The system is the submicroscopic reactant and product molecules.
  - The surroundings include the reaction solution itself and the calorimeter walls.

$$\begin{aligned} q_{rxn} &= -(q_{soln} + q_{cal}) \\ &= -(m_{soln}c_{soln}\Delta T_{soln} + C_{cal}\Delta T_{cal}) \end{aligned}$$



- Keep in mind that the measured solution is part of the surroundings and the system is considered closed. The measured ΔT reflects energy transfer to or from the surroundings—*NOT* the system of interest.
  - **Increase** in solution **temperature** → heat was released by the reaction to the surroundings and the reaction is **exothermic**.
  - **Decrease** in solution **temperature** → heat was absorbed by the reaction from the surroundings and the reaction is **endothermic**.

6.6. A 20.0 mL sample of 0.50 M NaOH and a 20.0 mL sample of 0.50 M HNO_3 are combined in a coffee-cup calorimeter. The temperature of the combined solutions increases from 20.4 °C to a maximum of 26.1 °C. Given that the heat capacity of the calorimeter is 25.2 J/°C, what is ΔH for the reaction in kJ/mol? You may assume that the density of the solution is 1.00 g/mL and the specific heat capacity is that of water (4.184 $J \cdot {}^{\circ}C^{-1} \cdot g^{-1}$).

Thermochemical Equations. Once the enthalpy of a reaction is known, it can be included in the chemical equation on the reactant side (for endothermic reactions) or the product side (for exothermic reactions). The result is a thermochemical equation, which we can use to carry out stoichiometric calculations involving heat. That is, we can answer questions about how much heat to expect given the scale of a reaction, or what scale is needed to release or absorb a certain amount of heat.

- **Thermochemical equation:** a balanced chemical equation that also includes the enthalpy change or **heat of reaction (ΔH_{rxn})** for 1 mole of reaction events of the equation as written.

$$C_6H_{12}O_6(s) + 6\ O_2(g) \rightarrow 6\ CO_2(g) + 6\ H_2O(g) + 2540\ kJ$$

$$C_6H_{12}O_6(s) + 6\ O_2(g) \rightarrow 6\ CO_2(g) + 6\ H_2O(g),\ \Delta H^{\circ} = -2540\ kJ$$

- The heat of reaction can be used in molar ratios for stoichiometric calculations ("How much heat…"). In other words, heat is now part of our stoichiometric toolbox!

6.7. The overall reaction in a commercial heat pack can be represented as follows.

$$\mathbf{4\ Fe(s) + 3\ O_2(g) \rightarrow 2\ Fe_2O_3(s),\ \Delta \mathit{H}^\circ = -1652\ kJ}$$

What is the change in enthalpy when 2.00 g of Fe(s) is reacted with excess oxygen?
Enter your response in kilojoules per mole with the appropriate sign.

6.C Thermochemistry III

Hess's Law (Fun with State Functions). The change in the value of a state function over a process does not depend on the intimate details of the process—only on the initial and final states. From this "fun fact" flows the idea that for a process that can be written as a sum of "sub-processes," the overall change in a state function is equal to the sum of the changes in that state function for each of the sub-processes. We will apply this idea to several different state functions before the end of the course. Here, we use it to calculate ΔH for unmeasured (or unmeasurable!) reactions that can be written as the sum of reactions with enthalpies we know.

- **Hess's law** is based on the idea that enthalpy is a state function.
  ΔH is the same regardless of the path followed to "get from" reactants to products.

- Consider a reaction R that can be written as a sum of reactions R_1 through R_N,

$$R = R_1 + R_2 + ... + R_N = \sum_i^N R_i$$

- The enthalpy change of reaction R is equal to the sum of the enthalpy changes for reactions R_1 through R_N,

$$\Delta H(R) = \Delta H(R_1) + \Delta H(R_2) + ... + \Delta H(R_N) = \sum_i^N \Delta H(R_i)$$

- Per Hess's law, for a reaction that can be written as the sum of multiple steps, the enthalpy is equal to the sum of the enthalpies of the individual steps.

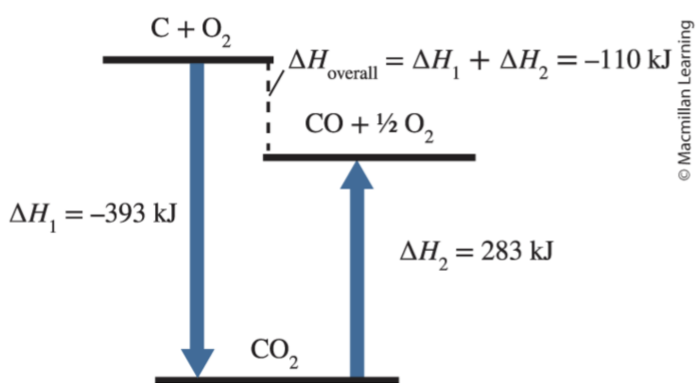


Figure 6.16. The enthalpy of reaction for oxidation of C to CO is $\Delta H_1 + \Delta H_2$.

- Per Hess's law, when a chemical equation is scaled or reversed, Δ*H* for the reaction is scaled or negated, respectively.
    - If you reverse ("negate") a reaction, change the sign of Δ*H*.

$$\Delta H(-R) = -\Delta H(R)$$

    - If you multiply a reaction by *n,* also multiply Δ*H* by *n*.

$$\Delta H(nR) = n\Delta H(R)$$

    - If you add two reactions, add their Δ*H* values.

$$\Delta H(R_1 + R_2) = \Delta H(R_1) + \Delta H(R_{@})$$

6.8. Use the information below to determine the standard enthalpy change for the reaction below in kilojoules (kJ).

$$B_2H_6(g) + 6\ Cl_2(g) \rightarrow 2\ BCl_3(g) + 6\ HCl(g)$$

$BCl_3(g) + 3\ H_2O(l) \rightarrow H_3BO_3(g) + 3\ HCl(g)$ $\Delta H^\circ = -112.5$ kJ

$\frac{1}{2}\ H_2(g) + \frac{1}{2}\ Cl_2(g) \rightarrow HCl(g)$ $\Delta H^\circ = -92.5$ kJ

$B_2H_6(g) + 6\ H_2O(l) \rightarrow 2\ H_3BO_3(g) + 6\ H_2(g)$ $\Delta H^\circ = -493.5$ kJ

Standard Enthalpy of Formation. The previous problem highlights how much of a pain enthalpy calculations can be when using a random smattering of chemical reactions. Establishing a standard process for the enthalpy "of" a substance makes these calculations *much* easier. The standard process is the formation of 1 mole of the compound from the elements in their standard states; the associated enthalpy is called enthalpy of formation. As any reaction can be thought of as decomposition of the reactants followed by formation of the products, enthalpies of formation can be used in a "products minus reactants" way to calculate standard enthalpies of reaction.

- **Standard enthalpy of formation (ΔH_f^o)** is the enthalpy change for the **formation reaction** of an element or compound occurring under standard conditions.
    - Formation is defined as: elements → 1 mol compound.

- Note that fractional stoichiometric coefficients are common on the reactant side of formation reactions.

- Because of the way formation is defined, ΔH_f^o for an element in its standard state is zero. E.g., $H_2(g) \rightarrow H_2(g)$ obviously has an enthalpy change of zero.

- Standard enthalpies of formation are *measured quantities* that will be provided in questions where needed. They can also be found in the Appendix of your textbook or in a large list here.

- Standard enthalpies of formation can be used to systematically calculate enthalpies of reaction. The key idea is thinking of the reaction as decomposition of the reactants—the reverse of formation—followed by formation of the products.

$$\text{reactants} \xrightarrow{-\Delta H_f^\circ} \text{elements} \xrightarrow{+\Delta H_f^\circ} \text{products}$$

- When the reaction involves multiple moles of reactants or products (nonzero coefficients), enthalpies of formation must be scaled accordingly.
    - Consider a general chemical equation,

$$r_1\,\mathrm{R}_1 + r_2\,\mathrm{R}_2 + \ldots + r_M\,\mathrm{R}_M \rightarrow p_1\,\mathrm{P}_1 + p_2\,\mathrm{P}_2 + \ldots + p_N\,\mathrm{P}_N$$

    - The standard enthalpy of reaction ΔH°_{rxn} can be written in terms of enthalpies of formation as "products minus reactants,"

$$\Delta H^\circ_{rxn} = \sum_j^N \Delta H^\circ_f(\mathrm{P}_j) - \sum_i^M \Delta H^\circ_f(\mathrm{R}_i)$$

Example. Calculate the standard enthalpy of reaction (ΔH°_{rxn}) for the reaction below from the given data.

$\Delta H_f^\circ[C_3H_8(g)]$ = –103.8 kJ
$\Delta H_f^\circ[CO_2(g)]$ = –395.3 kJ
$\Delta H_f^\circ[H_2O(l)]$ = –285.8 kJ

$$C_3H_8(g) \;+\; 5\,O_2(g) \;\rightarrow\; 3\,CO_2(g) \;+\; 4\,H_2O(l)$$

6.9. Use standard enthalpies of formation, available in Appendix A.1 of *Interactive General Chemistry* or below, to determine ΔH° for the following reaction in kilojoules per mole (kJ/mol).

$$P_4O_{10}(s) + 6\,H_2O(l) \rightarrow 4\,H_3PO_4(aq)$$

–2984 kJ –285.8 kJ –1271.7 kJ

Chapter Summary. Some of the amazing things we can now do with thermochemistry…

1. Define, apply, and carry out calculations related to pressure-volume work.

2. Define, apply, and carry out calculations related to heat.

3. Properly define system, surroundings, and universe when solving problems in thermodynamics; classify thermodynamic systems based on matter and energy transfer.

4. Use the first law of thermodynamics as a basis for energy balances.

5. Distinguish between state variables and path variables.

6. Apply the results of a calorimetry experiment to calculate an enthalpy change.

7. Apply Hess's law to calculate the enthalpy change of an unknown or unmeasured reaction using known enthalpy changes.

8. Apply standard enthalpies of formation to calculate standard enthalpy of reaction.